

LAVAL-ENTRAMMES AP, France

WMO: 071340

Lat: 48.0306N

Lon: 0.7464W

Elev: 97

StdP: 100.17

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.0	(c) -2.4	(d) -7.2	(e) 2.1	(f) -2.0	(g) -5.3	(h) 2.4	(i) -1.5	(j) 11.3	(k) 11.3	(l) 10.2	(m) 10.7	(n) 2.5	(o) 10	(p) 0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	30.6	20.2	28.2	19.2	26.2	18.5	21.2	28.2	20.1	26.2	19.2	24.7	3.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	13.8	23.5	18.0	13.1	22.4	17.1	12.4	21.6	61.8	28.2	57.9	26.3	55.0	24.8	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.2	(b) 8.1	(c) 7.2	(e) -6.2	(f) 34.1	(g) 2.2	(h) 2.1	(i) -7.8	(j) 35.6	(k) -9.1	(l) 36.8	(m) -10.3	(n) 38.0	(o) -12.0	(p) 39.5		
			DB													
			WB													
				-6.6	23.3	2.2	1.3	-8.2	24.2	-9.5	25.0	-10.8	25.7	-12.4	26.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	5.4	5.5	8.0	10.6	13.6	17.3	19.0	18.6	16.2	12.7	8.5	5.6		
	DBStd	5.92	3.67	3.51	3.17	3.08	3.05	3.27	2.97	2.88	2.90	3.23	3.16	3.62		
	HDD10.0	599	146	127	78	27	4	1	0	0	1	12	64	140		
	HDD18.3	2517	402	358	320	231	152	58	27	31	77	174	294	394		
	CDD10.0	1253	3	2	16	46	114	218	279	266	186	98	19	4		
	CDD18.3	129	0	0	0	0	3	26	48	38	12	1	0	0		
	CDH23.3	1252	0	0	0	8	41	274	462	350	113	4	0	0		
	CDH26.7	376	0	0	0	0	5	83	149	117	22	0	0	0		
(14) Wind	WSAvg	3.5	3.8	4.0	3.9	3.4	3.4	3.3	3.4	3.2	3.1	3.2	3.5	3.8		
(15) Precipitation	PrecAvg	801	79	66	55	58	66	51	57	52	56	84	81	89		
	PrecMax	1012	137	135	175	128	121	103	141	132	159	201	151	193		
	PrecMin	588	28	10	9	11	16	6	13	4	12	26	26	23		
	PrecStd	132	33	35	38	33	26	27	34	35	39	43	36	44		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.5	15.1	20.2	24.5	27.1	33.2	33.5	33.7	29.2	23.5	17.0	14.0		
		MCWB	12.1	10.9	12.9	16.0	18.7	22.2	21.1	21.5	18.5	17.5	14.6	12.6		
	2%	DB	12.6	12.9	17.2	21.7	24.1	28.9	30.5	30.0	26.5	20.7	15.6	13.1		
		MCWB	11.5	10.4	11.7	15.1	17.5	19.6	20.0	20.2	17.9	16.5	13.9	12.0		
	5%	DB	11.7	11.7	15.2	19.1	21.8	26.4	28.0	27.1	24.1	18.9	14.4	12.1		
		MCWB	10.6	9.9	11.1	13.6	16.0	18.9	19.1	19.0	17.4	15.6	12.9	11.1		
	10%	DB	10.5	10.6	13.2	16.7	19.7	24.1	25.8	24.7	22.0	17.5	13.3	11.1		
		MCWB	9.5	9.1	10.2	12.0	14.8	17.7	18.4	17.8	16.3	15.0	11.8	10.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.6	12.0	13.8	17.3	19.9	23.3	22.2	22.9	20.5	18.7	15.7	13.2		
		MCDB	13.1	13.4	18.3	22.3	25.4	30.1	31.3	30.8	25.8	21.8	16.3	13.7		
	2%	WB	11.7	11.1	12.5	15.4	18.0	20.8	20.9	21.1	19.2	17.2	14.1	12.2		
		MCDB	12.4	12.3	15.6	20.8	23.1	26.8	28.0	28.3	23.8	19.5	15.2	12.8		
	5%	WB	10.7	10.2	11.6	13.8	16.6	19.5	19.8	19.7	18.1	16.3	13.0	11.2		
		MCDB	11.5	11.4	14.1	18.2	21.0	25.2	26.2	25.3	22.7	18.2	14.2	12.0		
	10%	WB	9.7	9.2	10.7	12.4	15.3	18.3	19.0	18.6	17.1	15.5	12.0	10.2		
		MCDB	10.4	10.5	13.0	16.1	18.8	23.1	24.7	23.3	20.9	17.2	13.1	11.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	6.6	8.3	9.8	9.7	10.5	10.7	10.5	10.4	7.9	6.1	5.5		
		MCDBR	5.6	7.9	11.4	13.5	13.1	14.4	15.1	14.9	13.5	9.4	6.7	5.5		
		MCWBR	5.0	5.9	7.0	7.5	7.1	6.9	6.2	6.2	6.1	5.7	5.1	4.9		
	5% WB	MCDBR	5.3	7.1	9.7	12.6	11.9	13.4	13.2	12.7	11.2	8.0	6.1	5.5		
		MCWBR	5.1	5.6	6.4	7.3	6.8	7.1	6.0	5.9	5.8	5.3	5.0	5.2		
(40) Clear-Sky Solar Irradiance	taub		0.325	0.341	0.371	0.392	0.397	0.401	0.396	0.385	0.365	0.358	0.333	0.318		
	taud		2.426	2.383	2.294	2.238	2.271	2.295	2.309	2.360	2.394	2.417	2.430	2.400		
	Ebn at Noon		744	812	840	857	866	862	861	855	838	781	725	698		
	Edh at Noon		67	86	110	128	129	127	123	112	99	83	66	61		
(44) All-Sky Solar Radiation	RadAvg		1.02	1.85	2.95	4.40	5.20	5.78	5.51	4.83	3.81	2.20	1.26	0.88		
	RadStd		0.10	0.21	0.34	0.50	0.43	0.58	0.36	0.42	0.29	0.17	0.12	0.11		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page